

Armin Rezaiyan-Nojani

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EDUCATION

University of Maryland, College Park

Bachelor's of Science, Computer Science

College Park, MD

Jan 2024 — May 2026

- GPA: 3.3/4.0
- Clubs: Terps Racing
- Relevant Coursework: Computer Systems, Computer Networks & Security, Data Structures, Algorithms, Programming Language Theory, Artificial Intelligence, Data Science, Discrete Mathematics, Statistics & Probability, Calculus I/II, Linear Algebra

Thomas S. Wootton High School

High School Diploma

Rockville, MD

Sep 2019 — May 2023

- Varsity Soccer (4 years), Montgomery College Early College Program

WORK EXPERIENCE

Research and Development Engineer Intern

Auria

May 2025 — Present

Colorado Springs, CO

- Designed and deployed AI agents for satellite mission planning, enabling industry-specific workflows.
- Built a data pipeline with C# and Python to extract satellite data, enabling seamless ingestion by the agent.
- Led evaluation and adoption of AWS Bedrock, MCP integration, and driving production rollout.
- Co-authored a SmallSat 2025 publication on closed-loop tasking workflows with conversational agents.

Software Developer Intern

Vortex Vector

Jun 2024 — Sep 2024

Silver Spring, MD

- Built a hardware compatibility engine to automate component matching for custom glass assemblies.
- Designed a DynamoDB schema, unifying fragmented hardware catalogs across manufacturers.
- Utilize Python data parsers to extract, normalize, and filter product attributes for system integration.
- Accelerated design workflows by standardizing hardware selection and reducing manual lookup time for engineers.

Machine Learning Intern

NASA

Jun 2023 — Aug 2023

Greenbelt, MD

- Built a vector database and knowledge graph to improve discoverability of NASA's GES DISC archives.
- Linked PDF metadata and datasets using semantic embeddings for context-aware search.
- Integrated Neo4j relationships to enhance retrieval of related scientific materials.
- Benchmarked and published results at AGU 2023

Machine Learning Intern

NASA

Jun 2022 — Aug 2022

Greenbelt, MD

- Built an XGBoost Learning-to-Rank model to improve dataset relevance in NASA's GES DISC search engine
- Processed 230K+ user logs in Python to create training datasets and judgment lists.
- Engineered ranking features, boosting nDCG from 0.36 to 0.81.
- Delivered a prototype model and data pipeline adopted for future search-ranking research.

PUBLICATIONS

- [1] N. Dhingra, S. Kearns, N. Evans, E. Herz, and A. Rezaiyan-Nojani, "Closed-Loop Satellite Planning and Scheduling for Low-Latency Data Collection and Analysis with a Conversational Agent Interface," in *Small Satellite Conference*, Salt Lake City, UT, 2025. [Online]. Available: <https://digitalcommons.usu.edu/smallsat/2025/all2025/105/>
- [2] L. Kamper-Hinson, A. Rezaiyan-Nojani, and A. Mehrabian, "Enhancing Data and Knowledge Discoverability at NASA GES-DISC through Knowledge Graphs and Language Models," in *AGU Fall Meeting Abstracts*, 2023, pp. IN54A-1. [Online]. Available: <https://ui.adsabs.harvard.edu/abs/2023AGUFMIN54A..01K/abstract>

SKILLS

- Languages: Python, C++, C#, TypeScript, SQL, HTML/CSS, Assembly (MIPS, x86)
- Full-Stack/Cloud Development: React, Node.js, Express, MongoDB, PostgreSQL, REST APIs, Docker, AWS (S3, Lambda, DynamoDB)
- AI & Data Systems: Pandas, scikit-learn, NumPy, Pytorch, Neo4j, Weaviate/Pinecone, LangChain/LangGraph, Strands, AWS Bedrock